

Chapter 13 — FAIR data principles

Documentation supports humans

Learning objectives

- By the end of this part, learners should be able to name the four FAIR principles

What FAIR means

- The FAIR principles, Findable, Accessible, Interoperable
- Findable means datasets can be located through identifiers and rich metadata
- Accessible means retrieval under clear, standardized terms, which may include restrictions
- Interoperable means formats and vocabularies support combination with other datasets and
- Reusable means documentation, metadata

Example 13.12 — Indexing Genomics Data in Public Archives

- Example 13.12 — hands-on module
- Example 13.12 illustrates Findable sharing
- When a cancer genomics dataset is published
- Researchers can then locate it through keywords, metadata, or other relevant fields
- Indexing is what turns a deposited file into a discoverable research asset
- View files: ``modules/chapter13/example12/``

Example 13.13 — Zenodo and Figshare with Access Controls

- Example 13.13 — hands-on module
- Accessible does not mean every dataset must be free of restrictions
- Data must be available under clear terms
- Example 13.13 shows datasets hosted on Zenodo or Figshare
- Explore the chapter example module
- View files: ``modules/chapter13/example13/``

Example 13.14 — FASTQ VCF and Controlled Vocabularies

- Example 13.14 — hands-on module
- Interoperable data is structured for integration with other datasets and analytical tools
- Example 13.14 shows genomic data formatted in standard file types such as FASTQ or VCF so
- Controlled vocabularies such as Gene Ontology or the Unified Medical Language System
- Explore the chapter example module
- View files: ``modules/chapter13/example14/``

Example 13.15 — Community File Formats for Reuse

- Example 13.15 — hands-on module
- Reusable data is accompanied by documentation, metadata
- Example 13.15 prefers community formats such as CSV or JSON for tabular data
- Choosing formats the community already understands is a practical Reusable choice
- Explore the chapter example module
- View files: ``modules/chapter13/example15/``

Takeaways

- FAIR means Findable, Accessible, Interoperable, and Reusable for people and machines
- Public-archive indexing makes genomics and other deposits discoverable
- Accessibility allows clear terms and access controls, not only unrestricted dumps
- Standard formats and controlled vocabularies support interoperability
- Community file formats and licensing documentation enable reuse beyond the original study

Next

- Complete the quiz for this part
- The next part covers data-quality standards for reproducibility and research-facing